

Multi-stage events



1 $\frac{4}{9}$

2 a $\frac{1}{2}$

b $\frac{1}{2}$

c $\frac{219}{250}$

d $\frac{1}{10}$

e $\frac{321}{1000}$

3 a $\frac{1}{18}$

b $\frac{1}{36}$

4 a $\frac{1}{100}$
e $\frac{1}{5000}$

b $\frac{99}{100}$
f $\frac{4999}{5000}$

c $\frac{1}{10\,000}$

d $\frac{1}{5000}$

5 a $\frac{1}{25}$
e $\frac{7}{25}$

b 1

c 0

d $\frac{6}{25}$

6 a $\frac{1}{36}$

b $\frac{1}{36}$

c $\frac{1}{6}$

d $\frac{1}{36}$

7 a $\frac{5}{18}$

b $\frac{1}{6}$

c $\frac{7}{12}$

8 a $\frac{1}{36}$

b $\frac{1}{18}$

c $\frac{1}{12}$

d $\frac{1}{9}$



Tables

9

a

	2	3	5	8
1	2	3	5	8
2	4	6	10	16
3	6	9	15	24
4	8	12	20	32
5	10	15	25	40
6	12	18	30	48

b 24

c

i $\frac{1}{4}$ ii $\frac{1}{12}$ iii $\frac{2}{3}$

10

a

	1	2	3	4	5	6
1	1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
2	2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
3	3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
4	4, 1	4, 2	4, 3	4, 4	4, 5	4, 6
5	5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
6	6, 1	6, 2	6, 3	6, 4	6, 5	6, 6

b

i $\frac{1}{18}$ ii $\frac{1}{36}$ iii $\frac{1}{9}$ iv $\frac{1}{9}$
v $\frac{2}{3}$ vi $\frac{1}{6}$ vii $\frac{1}{18}$ viii $\frac{4}{9}$

c

i True ii False iii True iv False

11

a

	7	9	12
2	9	10	11
3	11	12	13
4	14	15	16



b 9

c

i $\frac{2}{9}$

ii $\frac{2}{9}$

iii $\frac{2}{9}$

12

a

	H	T
H	HH	HT
T	TH	TT

b $\frac{1}{4}$ c $\frac{3}{4}$

13

a

	1	2	3	4	5	6
H	H1	H2	H3	H4	H5	H6
T	T1	T2	T3	T4	T5	T6

b $\frac{1}{4}$ c $\frac{3}{4}$ d $\frac{1}{3}$

14

a

Spinner	A	B	C
1	1,A	1,B	1,C
2	2,A	2,B	2,C
3	3,A	3,B	3,C

b 9

c $\frac{2}{9}$ d $\frac{7}{9}$

15

a

	W	X	Y	Z
1	1,W	1,X	1,Y	1,Z
2	2,W	2,X	2,Y	2,Z
3	3,W	3,X	3,Y	3,Z
4	4,W	4,X	4,Y	4,Z
5	5,W	5,X	5,Y	5,Z
6	6,W	6,X	6,Y	6,Z



b 24

c $\frac{1}{8}$

d $\frac{1}{6}$

e $\frac{11}{12}$

Lists

16

a 5283, 5238, 5823, 5832, 5328, 5382, 3285, 3258, 3825, 3852, 3528, 3582

b 24

c $\frac{1}{4}$

d $\frac{1}{4}$

e $\frac{5}{12}$

17

a 689, 698, 869, 896, 968, 986

b 6

c i $\frac{1}{3}$ ii $\frac{1}{6}$ iii $\frac{1}{2}$

18

a 459, 495, 549, 594, 945, 954

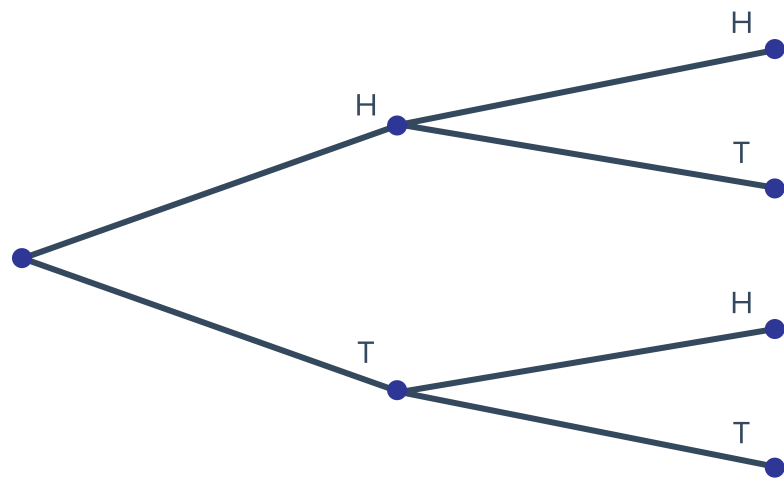
b i $\frac{2}{3}$ ii $\frac{1}{3}$ iii $\frac{2}{3}$ iv $\frac{1}{3}$



Tree diagrams

19

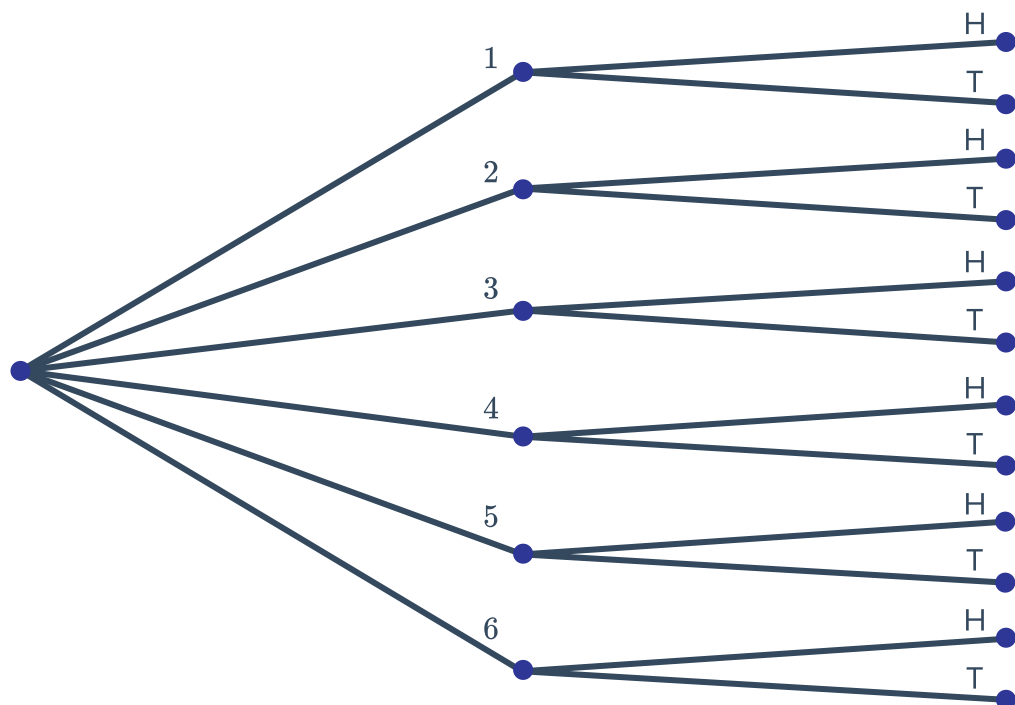
a



- b $\frac{1}{4}$
c $\frac{3}{4}$

20

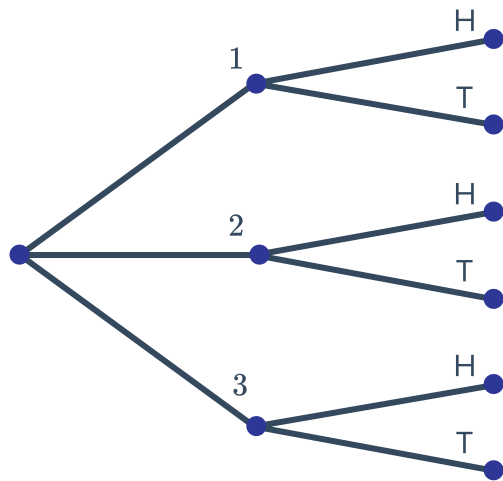
a



- b 1H, 1T, 2H, 2T, 3H, 3T, 4H, 4T, 5H, 5T, 6H, 6T
c $\frac{1}{2}$
d $\frac{5}{12}$

21

a



b

$$\frac{1}{6}$$



c

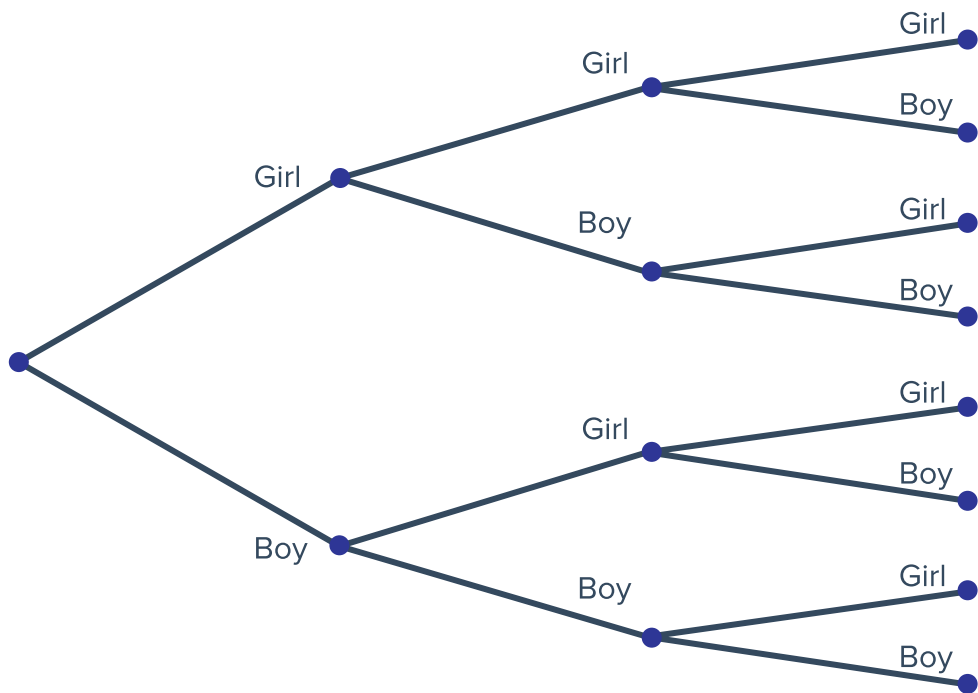
$$\frac{1}{2}$$

d

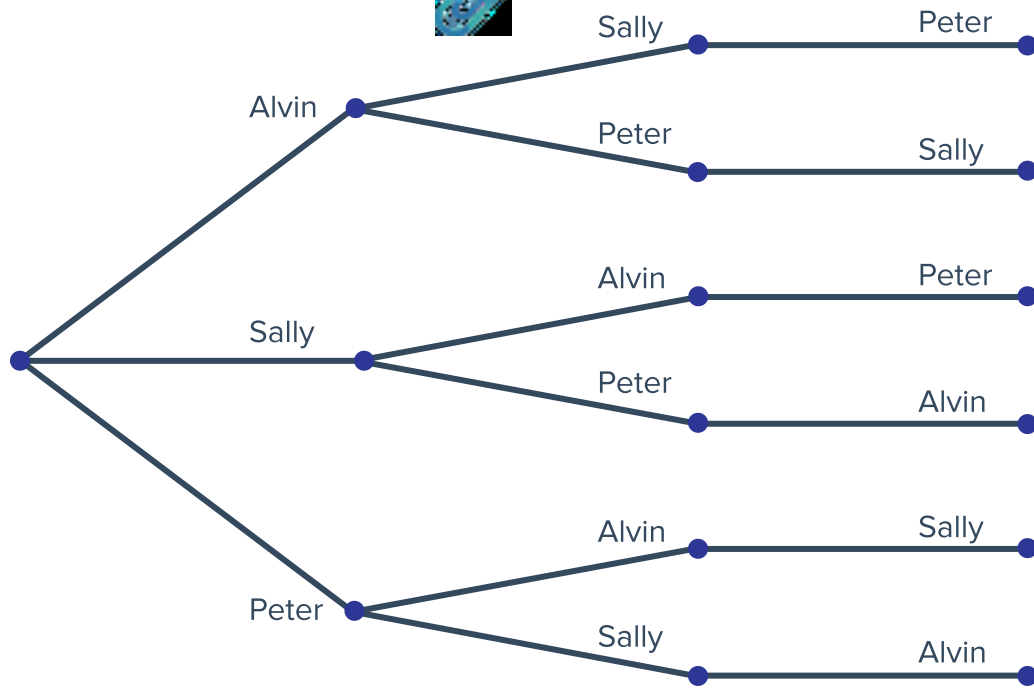
$$\frac{2}{3}$$

22

a

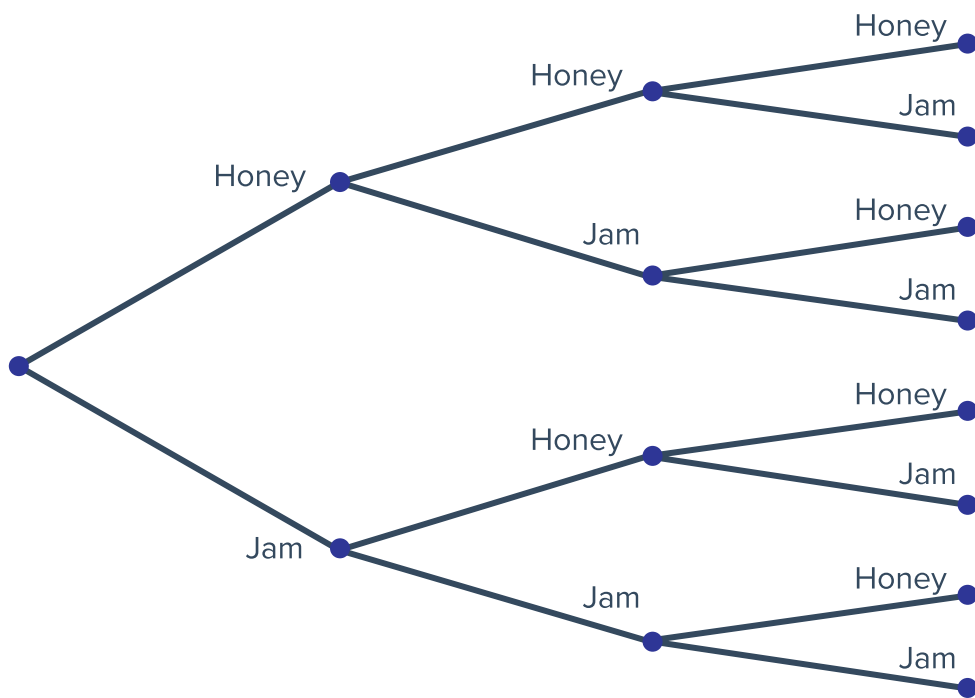


b



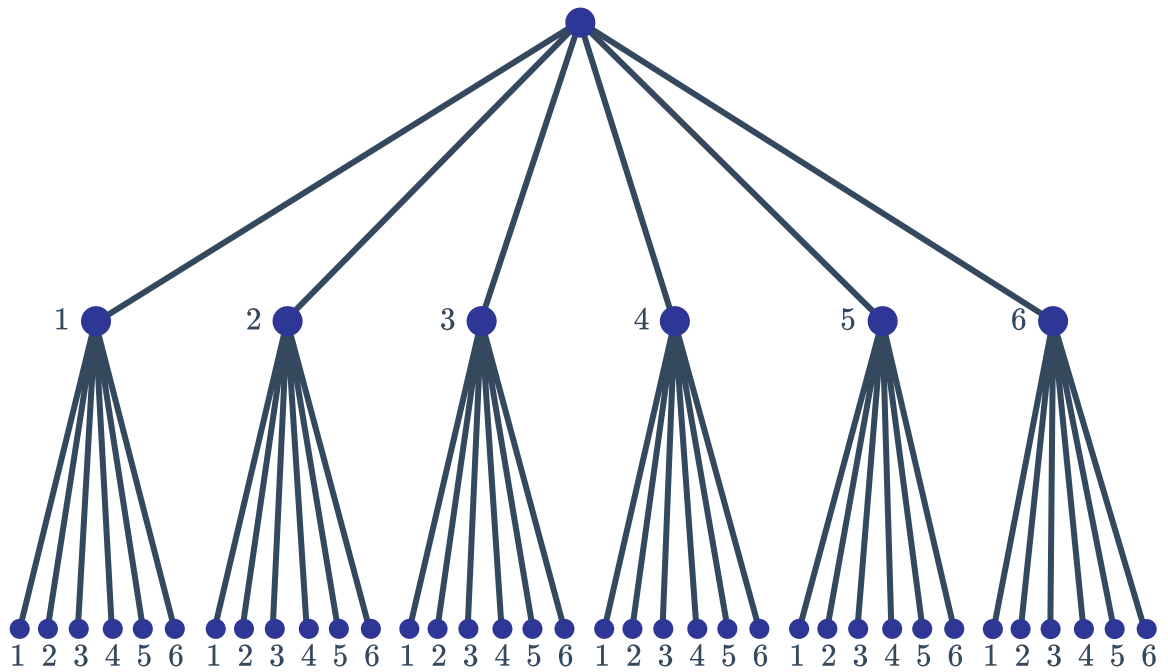
23

a



b $\frac{1}{2}$
c $\frac{1}{8}$

24
a



b

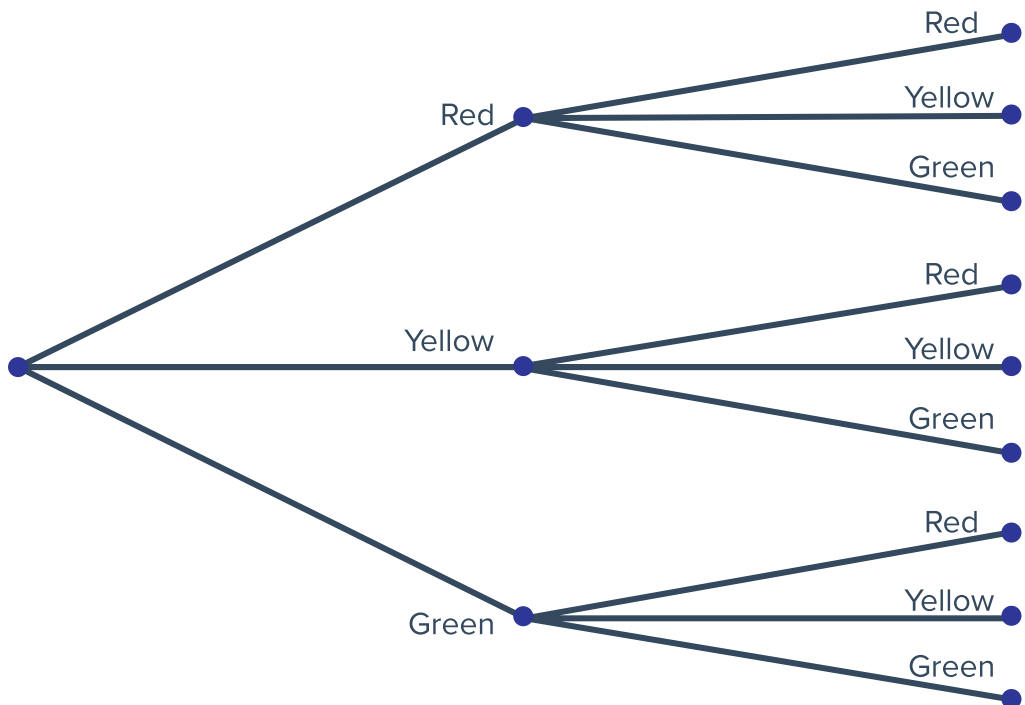
i $\frac{1}{36}$

ii $\frac{1}{6}$

iii $\frac{5}{6}$

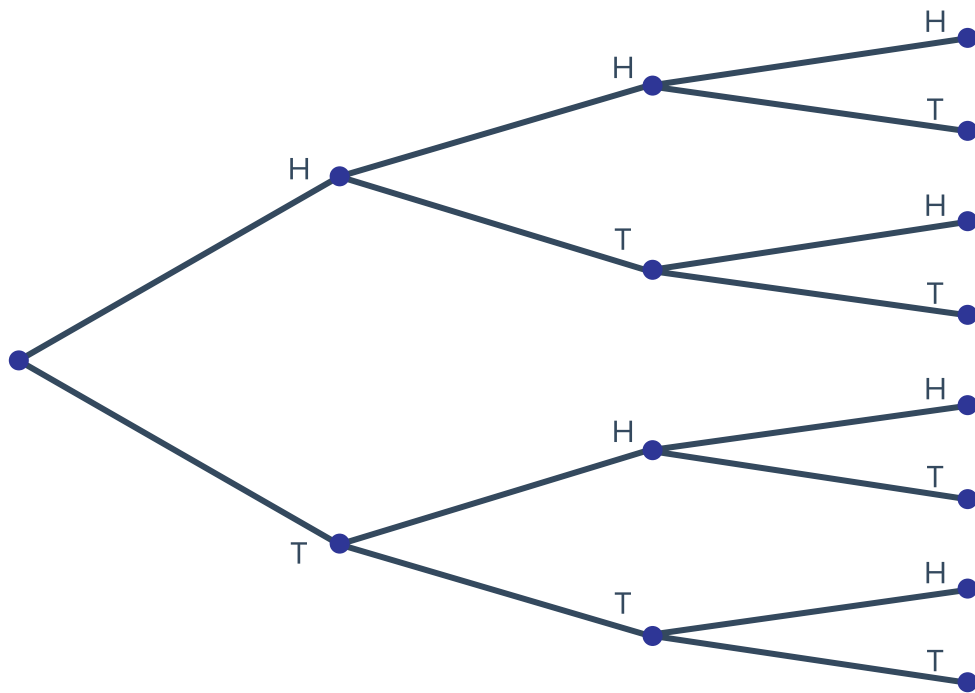
iv $\frac{1}{4}$

25
a



b $\frac{1}{9}$

26
a



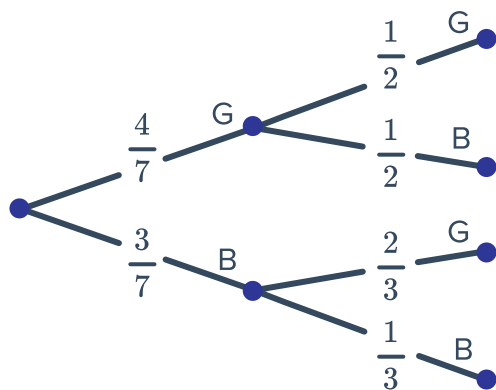
b

i $\frac{7}{8}$
iii $\frac{1}{8}$

ii $\frac{1}{8}$

Probability trees

27
a

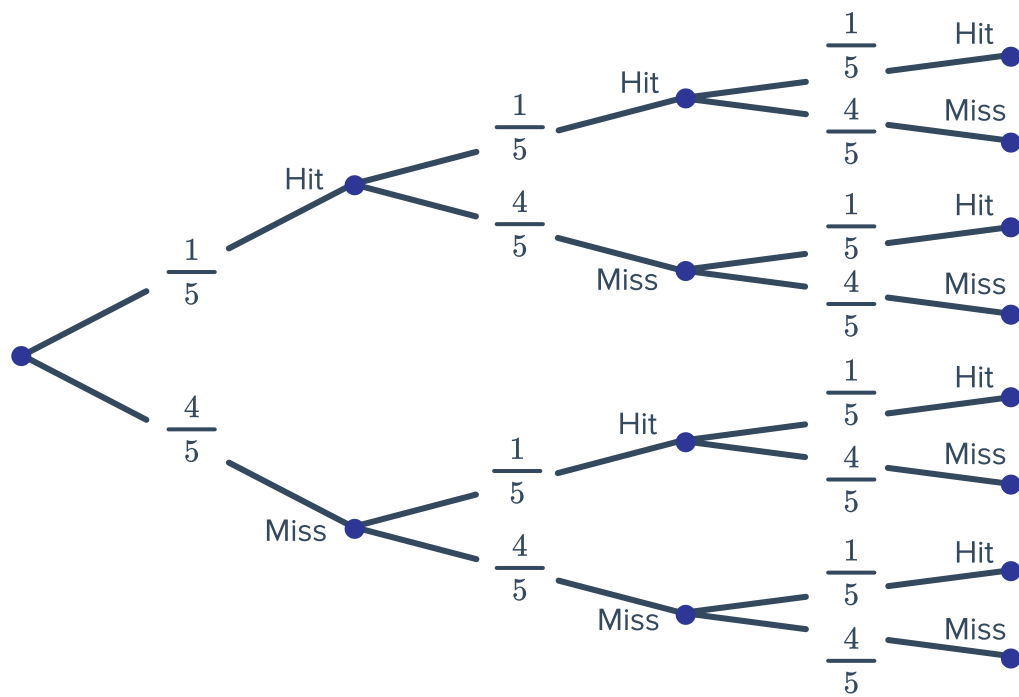


b $\frac{3}{7}$

c $\frac{2}{7}$

28

a



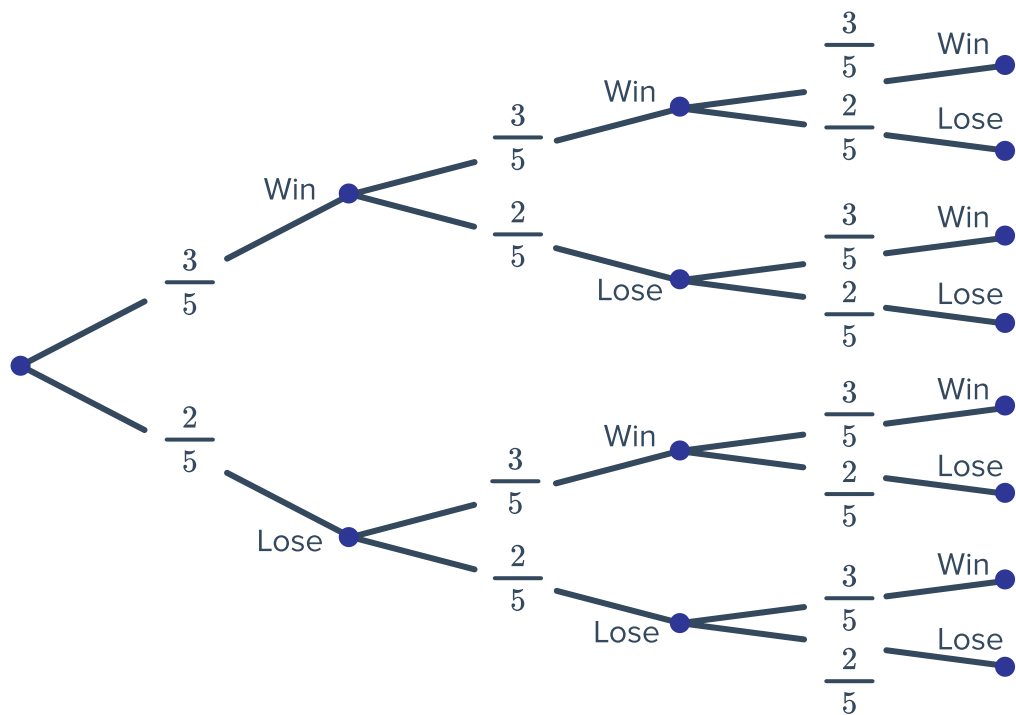
b $\frac{1}{125}$

c $\frac{124}{125}$

d $\frac{61}{125}$

29

a

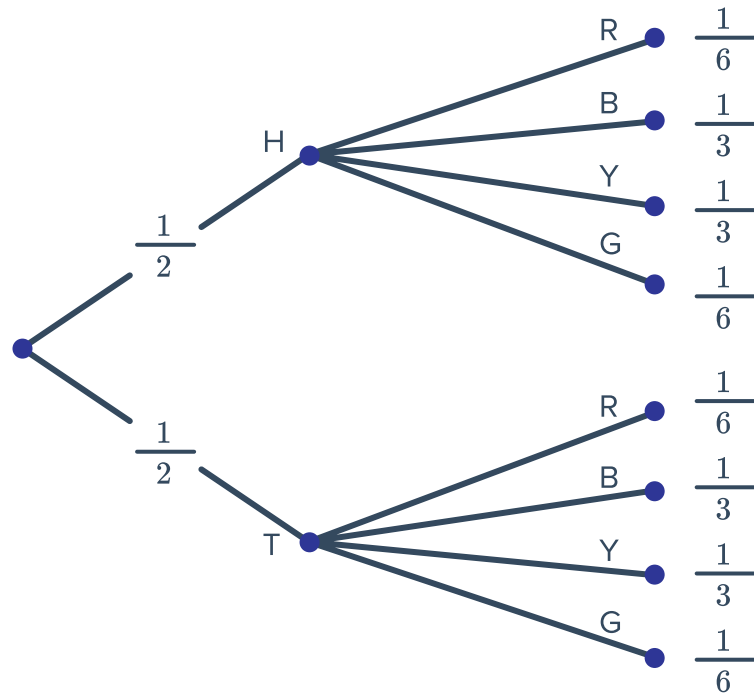


b $\frac{27}{125}$

- c $\frac{8}{125}$
- d $\frac{81}{125}$

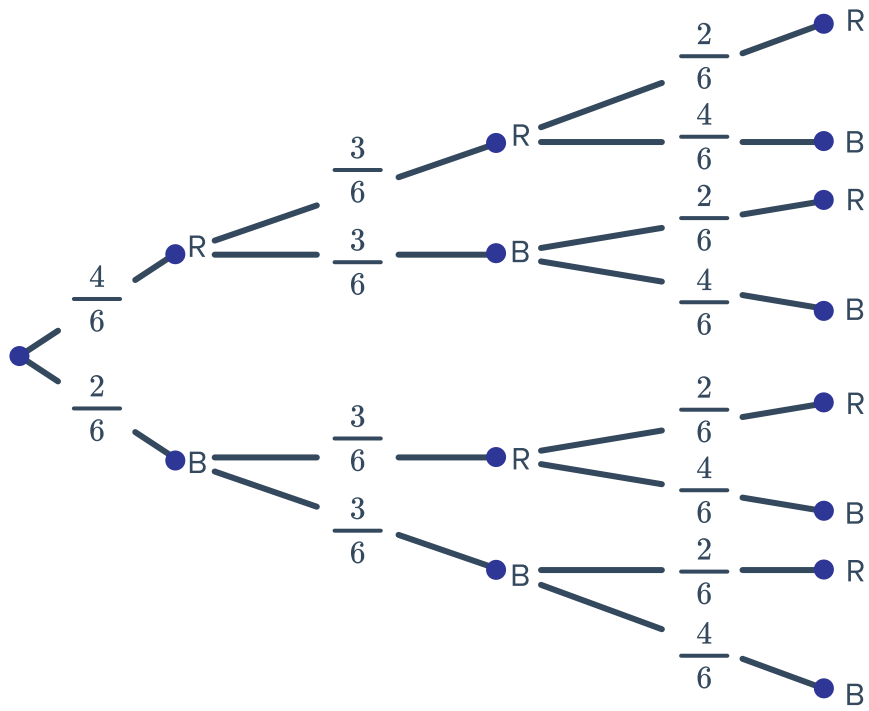


30
a



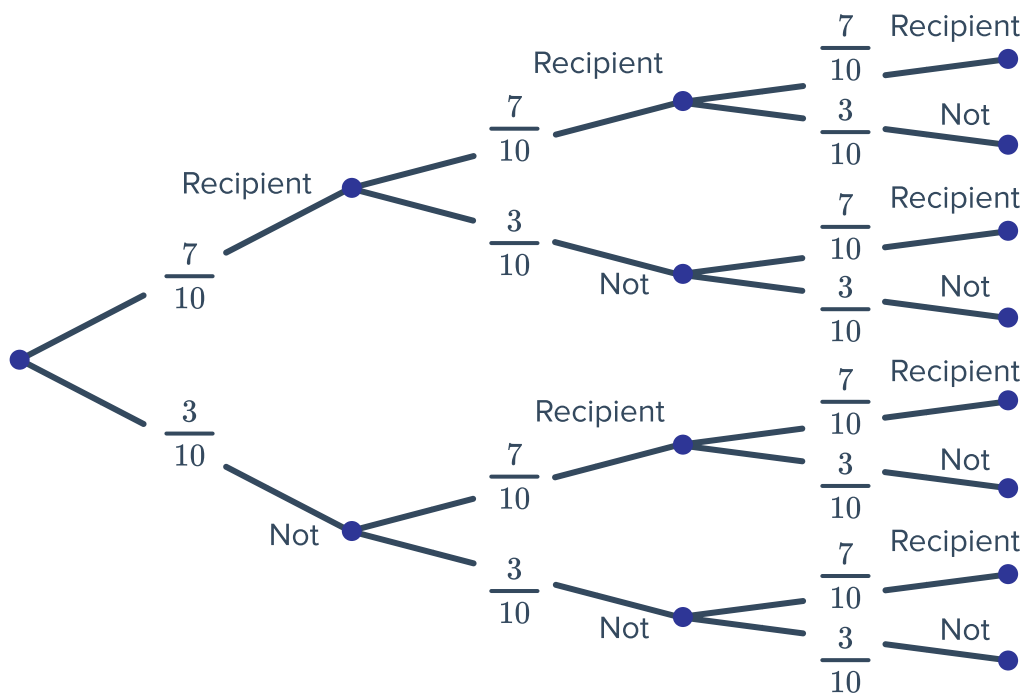
- b $\frac{1}{6}$
- c $\frac{2}{3}$
- d $\frac{5}{12}$
- e $\frac{5}{12}$

31
a



- b $\frac{1}{9}$
- c $\frac{1}{2}$
- d $\frac{7}{18}$

32
a



- b $\frac{973}{1000}$

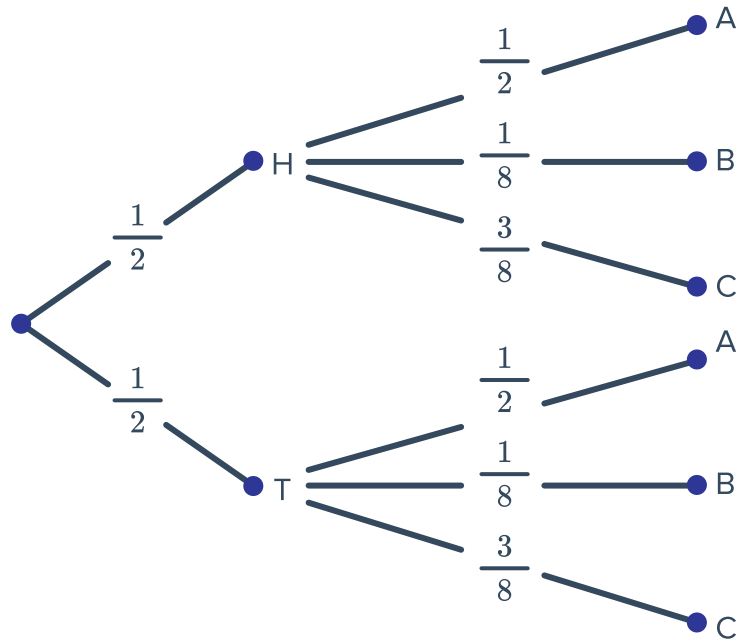
c $\frac{657}{1000}$

d $\frac{63}{100}$



33

a



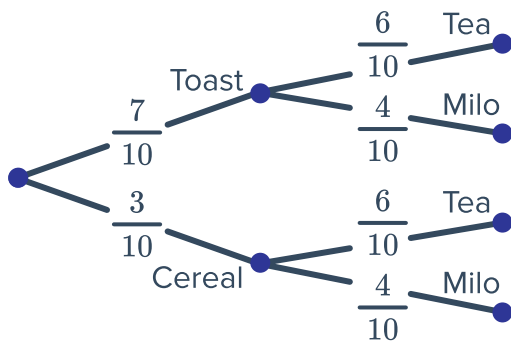
b $\frac{1}{4}$

c $\frac{3}{4}$

34

a

b 0.42



c 0.88